

6. (a) The table shows the names, molecular formulae and structural formulae of some alkanes.

(i) Complete the table.

[3]

Name	Molecular formula	Structural formula
.....	CH_4	$ \begin{array}{c} \text{H} \\ \\ \text{H} - \text{C} - \text{H} \\ \\ \text{H} \end{array} $
ethane	C_2H_6	
propane		$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & \text{H} & \\ & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - \text{H} \\ & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & \end{array} $
butane	C_4H_{10}	$ \begin{array}{cccccccc} & \text{H} & & \text{H} & & \text{H} & & \text{H} \\ & & & & & & & \\ \text{H} & - \text{C} & - & \text{C} & - & \text{C} & - & \text{C} - \text{H} \\ & & & & & & & \\ & \text{H} & & \text{H} & & \text{H} & & \text{H} \end{array} $

- (ii) Give the **names** of the **two** elements present in all alkanes.

[1]

..... and

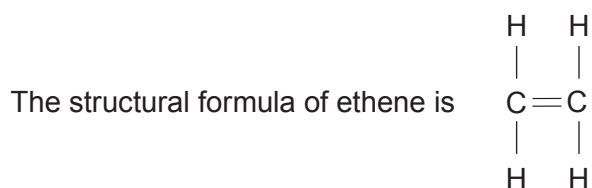
- (iii) The general formula for the alkanes is $\text{C}_n\text{H}_{2n+2}$.

Circle the molecular formula for the alkane containing 6 carbon atoms.

[1]

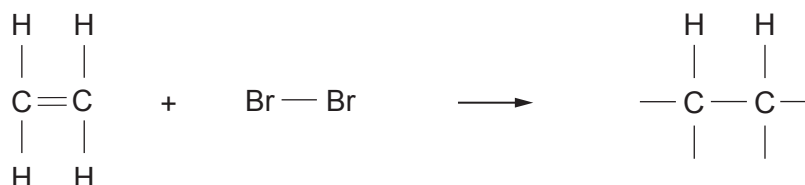


- (b) Ethene belongs to a different family of hydrocarbons called the alkenes.



The presence of the double bond in alkenes can be identified using bromine water.

- (i) Complete the structural formula for the product of the reaction between ethene and bromine. [1]



- (ii) Complete the sentence below by underlining the correct words in the brackets. [2]

Ethene turns (**purple / orange / green / colourless**) bromine water
(**green / milky / colourless / blue**).

- (c) Underline the name given to the type of reaction taking place when ethene reacts with bromine. [1]

addition

polymerisation

oxidation

neutralisation

